Michael Galileo Hasian (201344020)	**Rangkuman Minggu ke-11**	Senin, 11 November 2022
		Aplikasi Mikrokontroler
3-TNK		Ferry Satria, BSEE, M.T. Rahmawati Hasanah, S.ST.,M.T.

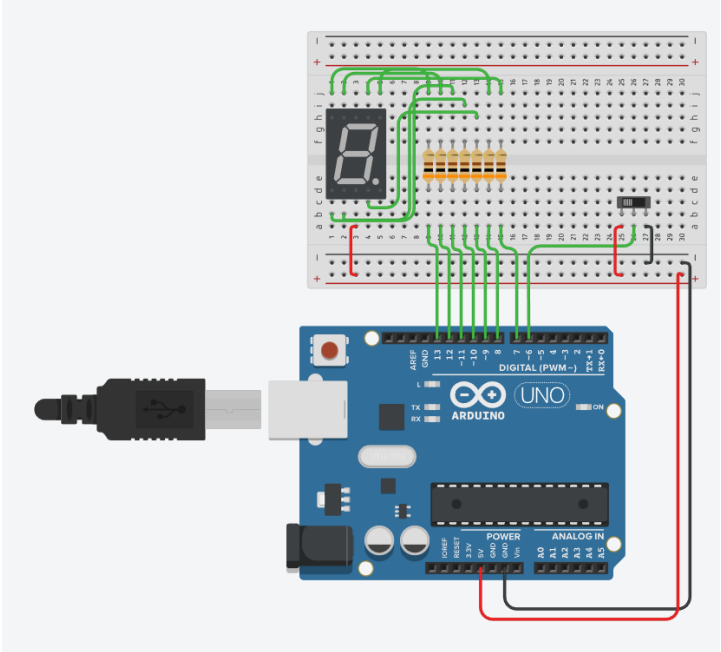
Tabel Kode Heksadesimal untuk 7-Segment Display

Angka	Common Anode	Common Cathode
0	C0	3F
1	F9	06
2	A4	5B
3	B0	4F
4	99	66
5	92	6D
6	82	7D
7	F8	07
8	80	7F
9	90	6F
A	88	77
B	83	7C
C	C6	39
D	A1	5E
E	86	79
F	8E	71

Buat Program untuk membuat Up Down Counter yang menghitung 0-F

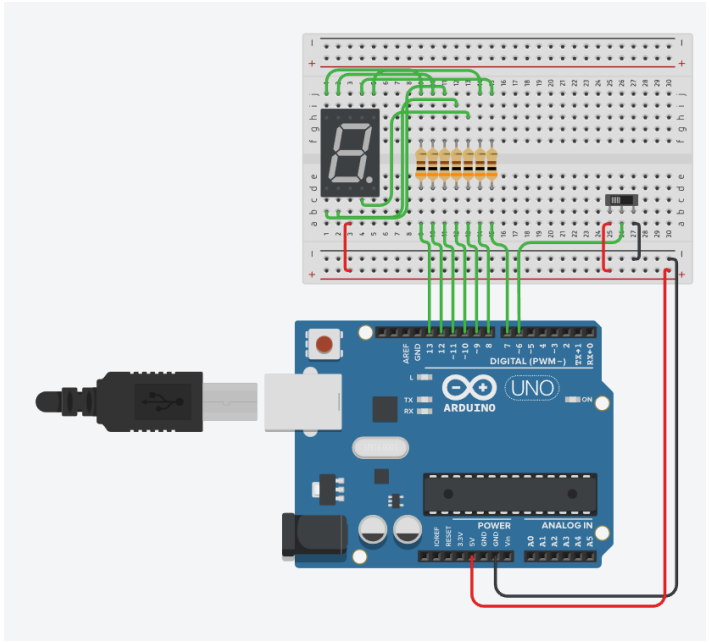
```
int pola[]={0XC0, 0X79, 0X24, 0XB0, 0X99, 0X12, 0X82, 0XF8, 0X80, 0X98, 0X88,
0X83, 0XC6, 0XA1, 0X86, 0X8E };
int i,x;
int pin[7]={7,8,9,10,11,12,13};
int pos[7]={1,2,4,8,0X10,0X20,0X40};
int mode,j=0;
void setup()
```

```
{ pinMode(6,INPUT);
 for(i=0; i<7; i++)
   pinMode(pin[i], OUTPUT);}
void display(int data)
{ x = pola[data];
   for(i=0; i<7; i++)
   digitalWrite(pin[i], x & pos[i]);}
void loop()
   { mode = digitalRead(6);
    if(mode==HIGH)
    {j++; if(j>15) {j=0;}}
    else
    {j--; if(j<0) {j=15;}
    }
    display(j);
    delay(500);
   }
```



Buat program untuk membuat Up Down Counter Mod 10

```
int pola[]={0XC0, 0X79, 0X24, 0XB0, 0X99, 0X12, 0X82, 0XF8, 0X80, 0X98};
int i,x;
int pin[7]={7,8,9,10,11,12,13};
int pos[7]={1,2,4,8,0X10,0X20,0X40};
int mode,j=0;
void setup()
{ pinMode(6,INPUT);
 for(i=0; i<7; i++)
   pinMode(pin[i], OUTPUT);}
void display(int data)
{ x = pola[data];
   for(i=0; i<7; i++)
   digitalWrite(pin[i], x & pos[i]);}
void loop()
   { mode = digitalRead(6);
    if(mode==HIGH)
    {j++; if(j>15) {j=0;}}
    else
    {j--; if(j<0) {j=15;}
    }
    display(j);
    delay(700);
   }
```


DIGITAL (PWM~)
UNO
ARDUINO
POWER
ANALOG IN

DOT MATRIK LAB

```
.INCLUDE "M8535DEF.INC"
.ORG 0X00
.EQU TIMER_1=0XFF98
RJMP MAIN

.ORG 0X80
DAT:
.DB 0X30,0X36,0X36,0X36,0X6,0
.DB 0X10,8,4,2,1

MAIN:       LDI R16,LOW (RAMEND)
            OUT SPL, R16
            LDI R16, HIGH (RAMEND)
            OUT SPH, R16

            LDI R16,0XFF
            OUT DDRA,R16
            OUT DDRB,R16

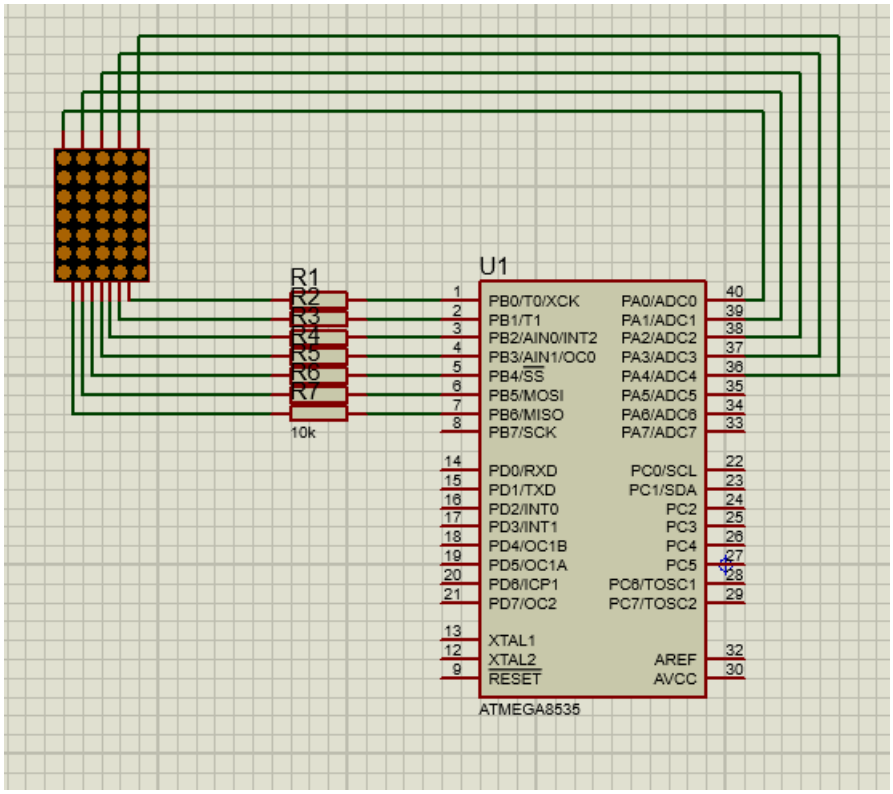
            LDI R17,6
LL0:  LDI R18,5
            LDI R30,LOW(2*DAT)
            LDI R31,HIGH(2*DAT)

LL1:  LPM R16,Z
            OUT PORTB,R16
            ADD R30,R17

            LPM R16,Z
            OUT PORTA,R16
            RCALL D_2MS
            SUB R30,R17
```

```
                INC R30
                DEC R18
                BRNE LL1
                RJMP LL0

;----------SUBRUTIN DELAY 2MS-----------
D_2MS:          PUSH R16
                PUSH R17
                LDI R16,1
                OUT TIMSK,R16
                LDI R16, HIGH(TIMER_1)
                OUT TCNT1H,R16
                LDI R16, LOW(TIMER_1)
                OUT TCNT1L,R16
                LDI R16,5
                OUT TCCR1B,R16
LT:             IN R17,TIFR
                SBRS R17,TOV1
                RJMP LT
                LDI R16,4
                OUT TIFR,R16
                POP R17
                POP R16
                RET
```



LCD

```
#include <LiquidCrystal.h>

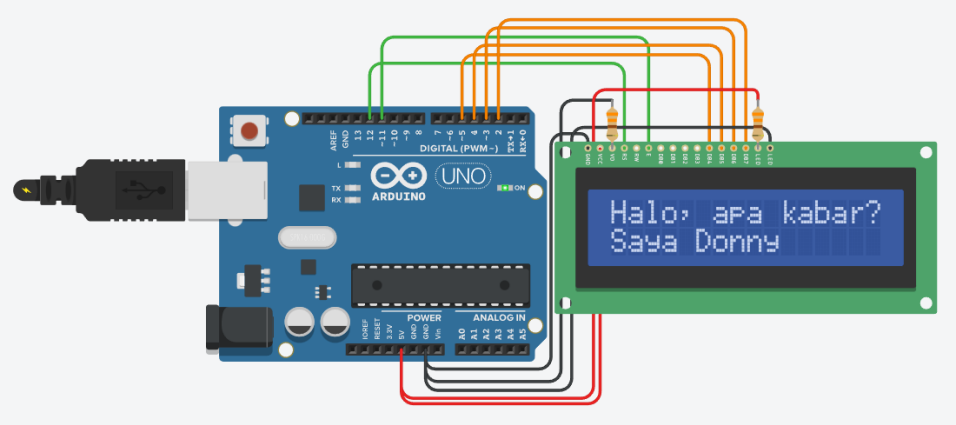

const int rs=12,en=11,d4=5, d5=4, d6=3,d7=2;
LiquidCrystal lcd(rs,en,d4,d5,d6,d7); // mesign nilai yg ada diatasnya
int i,x=0; // inisialisasi nilai i dan x
void setup()
{
 lcd.begin(16,2); //
 lcd.setCursor(0,0);
 lcd.print("Halo, apa kabar?"); //menampulkan huruf atau angka
  lcd.setCursor(0,1);
  lcd.print("Saya Donny");delay(4000); // mencetak huruf atau angka, ada delaynya

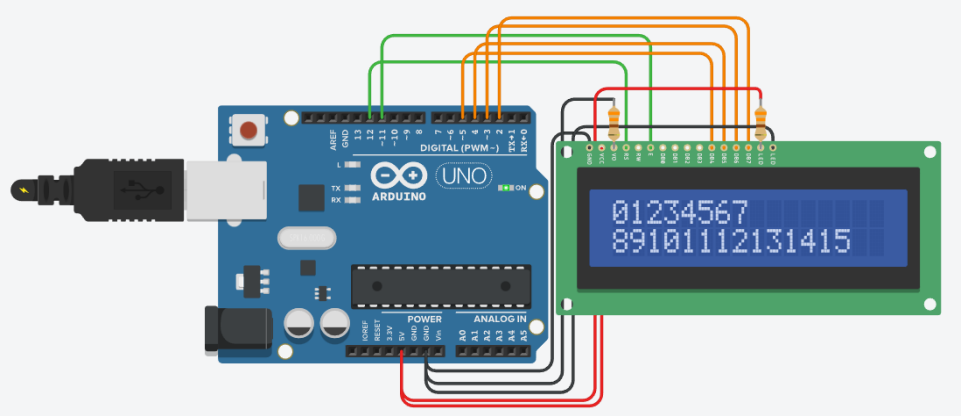

}


void loop()
{ lcd.clear(); //menghapus
 lcd.setCursor(0,0);  // menset ada di baris 1, nanti berpindah kebaris yang lainnya
```

```
for(i=x; i<=x+7; i++) // nyetak 2 digit 2 digit
 {if(i>99)break; //
  lcd.print(i); // mencetak huruf atau angka
  delay(700);
 }
lcd.setCursor(0,1);
for(i=x+8; i<=x+15; i++) // nyetak 1 digit 1 digit
 {if(i>99)break;

  lcd.print(i);
  delay(700);
 }
 x=i;
if(x>99)x=0;
}
```





Soal Lab

1. Membuat program menampilkan kata “BISA” bergerak bolak balik (kekiri dan kekanan setiap 5 kali) dari baris 1 ke baris 2 secara berulang. Tentukan sendiri kecepatan pergerakannya.

- **Kode Program**

```
#include <LiquidCrystal.h>

const int rs=12,en=11,d4=5, d5=4, d6=3,d7=2;
LiquidCrystal lcd(rs,en,d4,d5,d6,d7);
int i,x=0;
void setup()
{
 lcd.begin(16,2);
}

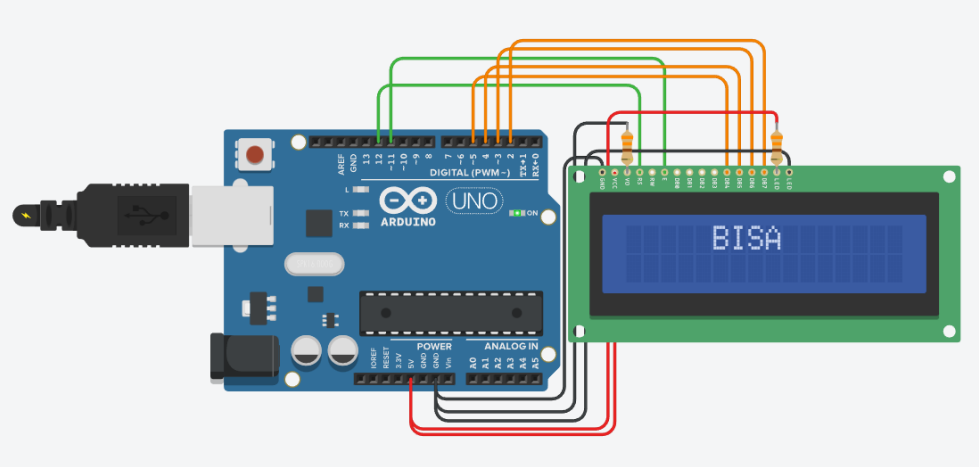
void loop() {
lcd.setCursor(0,0);
lcd.print("BISA");
 for (i=0 ; i<=11; i++) {
        lcd.scrollDisplayRight();
        delay(50);
        }
 for (i=0 ; i<=11; i++) {
        lcd.scrollDisplayLeft();
        delay(50);
        }
 for (i=0 ; i<=11; i++) {
        lcd.scrollDisplayRight();
        delay(50);
        }
 for (i=0 ; i<=11; i++) {
        lcd.scrollDisplayLeft();
        delay(50);
```

```
            }
  for (i=0 ; i<=11; i++) {
            lcd.scrollDisplayRight();
            delay(50);
            }
  for (i=0 ; i<=11; i++) {
            lcd.scrollDisplayLeft();
            delay(50);
            }
  for (i=0 ; i<=11; i++) {
            lcd.scrollDisplayRight();
            delay(50);
            }
  for (i=0 ; i<=11; i++) {
            lcd.scrollDisplayLeft();
            delay(50);
            }
  for (i=0 ; i<=11; i++) {
            lcd.scrollDisplayRight();
            delay(50);
            }
  for (i=0 ; i<=11; i++) {
            lcd.scrollDisplayLeft();
            delay(50);
            }
lcd.clear();
lcd.setCursor(0,1);
lcd.print("BISA");
  for (i=0 ; i<=11; i++) {
            lcd.scrollDisplayRight();
            delay(50);
            }
  for (i=0 ; i<=11; i++) {
```

```
        lcd.scrollDisplayLeft();
        delay(50);
        }
    for (i=0 ; i<=11; i++) {
        lcd.scrollDisplayRight();
        delay(50);
        }
    for (i=0 ; i<=11; i++) {
        lcd.scrollDisplayLeft();
        delay(50);
        }
    for (i=0 ; i<=11; i++) {
        lcd.scrollDisplayRight();
        delay(50);
        }
    for (i=0 ; i<=11; i++) {
        lcd.scrollDisplayLeft();
        delay(50);
        }
    for (i=0 ; i<=11; i++) {
        lcd.scrollDisplayRight();
        delay(50);
        }
    for (i=0 ; i<=11; i++) {
        lcd.scrollDisplayLeft();
        delay(50);
        }
    for (i=0 ; i<=11; i++) {
        lcd.scrollDisplayRight();
        delay(50);
        }
    for (i=0 ; i<=11; i++) {
        lcd.scrollDisplayLeft();
```

```
            delay(50);
            }
      lcd.clear();
      }
```

- **Ouput Program**



2. Membuat 4 counter (menghitung naik secara berulang), Tentukan sendiri kecepatan pergerakannya
   - Ujung kiri atas up counter desimal mod 10
   - Ujung kanan atas up counter desimal mod 100
   - Ujung kiri bawah down counter HD mod 16
   - Ujung kanan bawah down counter HD mod 256 (FF-00)

3. Membuat program seperti soal no 1, selain bergerak tapi berkedip juga

- **Kode Program**

```
 #include <LiquidCrystal.h>

const int rs=12,en=11,d4=5, d5=4, d6=3,d7=2;
LiquidCrystal lcd(rs,en,d4,d5,d6,d7);
int i,x=0;

void setup() {
  lcd.begin(16, 2);
}
```

```
void loop() {
  lcd.setCursor(0,0);
  lcd.print("BISA");

  for (int positionCounter = 0; positionCounter < 12; positionCounter++)
{
    lcd.scrollDisplayRight();
    delay(50);
    lcd.noDisplay();
    delay(50);
    lcd.display();
    delay(50);
  }

  for (int positionCounter = 0; positionCounter < 12; positionCounter++)
{
    lcd.scrollDisplayLeft();
    delay(50);
    lcd.noDisplay();
    delay(50);
    lcd.display();
    delay(50);
  }

  for (int positionCounter = 0; positionCounter < 12; positionCounter++)
{
    lcd.scrollDisplayRight();
    delay(50);
    lcd.noDisplay();
    delay(50);
    lcd.display();
    delay(50);
```

```
  }

  for (int positionCounter = 0; positionCounter < 12; positionCounter++)
{
    lcd.scrollDisplayLeft();
    delay(50);
    lcd.noDisplay();
    delay(50);
    lcd.display();
    delay(50);
  }

  for (int positionCounter = 0; positionCounter < 12; positionCounter++)
{
    lcd.scrollDisplayRight();
    delay(50);
    lcd.noDisplay();
    delay(50);
    lcd.display();
    delay(50);
  }

  lcd.clear();

  lcd.setCursor(0,1);
  lcd.print("BISA");

  for (int positionCounter = 0; positionCounter < 12; positionCounter++)
{
    lcd.scrollDisplayRight();
    delay(50);
    lcd.noDisplay();
    delay(50);
```

```
    lcd.display();
    delay(50);
  }

  for (int positionCounter = 0; positionCounter < 12; positionCounter++)
 {
    lcd.scrollDisplayLeft();
    delay(50);
    lcd.noDisplay();
    delay(50);
    lcd.display();
    delay(50);
  }

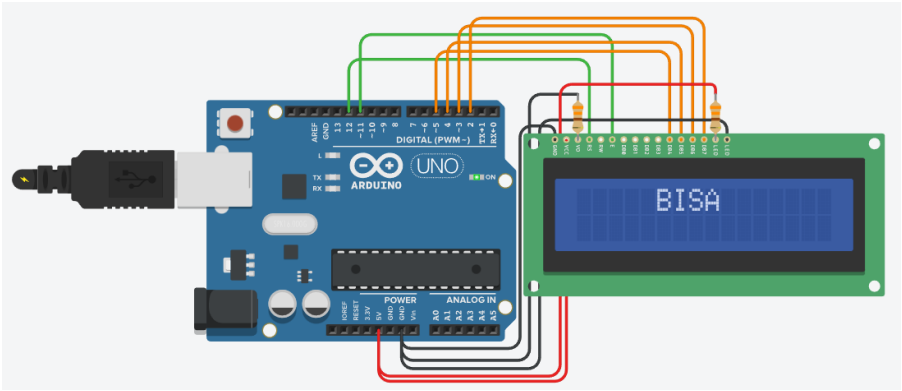
  for (int positionCounter = 0; positionCounter < 12; positionCounter++)
 {
    lcd.scrollDisplayRight();
    delay(50);
    lcd.noDisplay();
    delay(50);
    lcd.display();
    delay(50);
  }

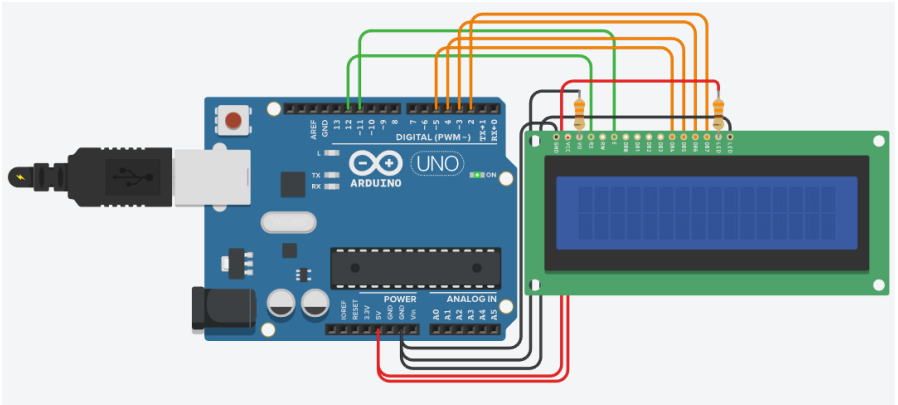
  for (int positionCounter = 0; positionCounter < 12; positionCounter++)
 {
    lcd.scrollDisplayLeft();
    delay(50);
    lcd.noDisplay();
    delay(50);
    lcd.display();
    delay(50);
  }
```

```
        for (int positionCounter = 0; positionCounter < 12; positionCounter++)
      {
        lcd.scrollDisplayRight();
        delay(50);
        lcd.noDisplay();
        delay(50);
        lcd.display();
        delay(50);
       }

        lcd.clear();
        delay(150);
}
```

- **Output Program**





4. Membuat program UP/Down counter desimal mod 100 (00-99) yang mode up/downnya dikendalikan secara polling (pilih sendiri) pin untuk menghubungkan push button pollingnya.